

A Survey of Adversarial Defenses and Robustness in NLP

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In the past few years, it has become increasingly evident that deep neural networks are not resilient enough to withstand adversarial perturbations in input data, leaving them vulnerable to attack. Various authors have proposed strong adversarial attacks for computer vision and Natural Language Processing (NLP) tasks. As a response, many defense mechanisms have also been proposed to prevent these networks from failing. The significance of defending neural networks against adversarial attacks lies in ensuring that the model's predictions remain unchanged even if the input data is perturbed. Several methods for adversarial defense in NLP have been proposed, catering to different NLP tasks such as text classification, named entity recognition, and natural language inference. Some of these methods not only defend neural networks against adversarial attacks but also act as a regularization mechanism during training, saving the model from overfitting. This survey aims to review the various methods proposed for adversarial defenses in NLP over the past few years by introducing a novel taxonomy. The survey also highlights the fragility of advanced deep neural networks in NLP and the challenges involved in defending them.

CCS Concepts: • **Computer systems organization** → **Embedded systems**; *Redundancy*; Robotics; • **Networks** → Network reliability;

Additional Key Words and Phrases: Adversarial attacks, adversarial defenses, perturbations, NLP

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1 INTRODUCTION

Recently, there have been significant advancements in the field of **Natural Language Processing (NLP)** using deep learning algorithms. In fact, the proposed solutions for NLP have already surpassed human accuracy in some cases [27, 52, 60]. By learning from vast amounts of available data, deep learning has revolutionized the field by providing a representation of language that can be utilized for a range of tasks. NLP involves the manipulation and processing of human language, and deep neural networks have enabled NLP models to learn how to represent language and solve tasks such as text classification, natural language inferencing, sentiment analysis, machine translation, named entity recognition, malware detection, reading comprehension, textual

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entailment with remarkable accuracy [52, 83, 108]. A typical NLP pipeline using deep neural networks learns word representation in text and contextual details in sentences, and this type of language modeling can be utilized for tasks such as sentence classification, translation, and question answering, using **Convolutional Neural Network (CNN)** or **Recurrent Neural Networks (RNN)**-based learning models. Finding feature representations for natural language text is a crucial part of this pipeline, and various methods have been proposed, including hand-crafted features and auto-encoded features using recent deep neural networks [59, 83, 108].

Despite the significant progress made, deep neural networks still suffer from a lack of interpretability and operate as a “black box” [12]. Their high performance remains inexplicable, and there is limited understanding of how they function [12, 96]. Although they can achieve exceptional accuracy and human-like performance, these networks are vulnerable to attacks and are highly sensitive to even the slightest perturbations in inputs, causing them to fail [1, 27]. Recently, there has been a growing number of proposed adversarial attacks for deep neural networks in computer vision and NLP, raising concerns about the robustness of these high-performing models [137, 145]. Adversarial attacks can pose a significant security threat to applications such as malware and spam detection, as well as biometrics. In NLP, these attacks can take various forms, including substitution, insertion, deletion, and swapping of words/characters in a sentence or finding a neighboring word embedding to introduce perturbations in the input [145].

There are two main types of adversarial attacks, black box and white box, based on the attacker’s access to the model’s parameters [145]. These attacks can be further categorized based on their design granularity, including character-level, word-level, sentence-level, and multi-level attacks [39, 117, 145]. Adversaries are generated by perturbing the input text using techniques such as insertion, deletion, flipping, swapping of characters or words, or paraphrasing the sentence in a way that preserves its original meaning but changes the wording. In white-box attacks, the attacker has access to the model’s parameters and modifies the word embeddings of input text using gradient-based schemes. In contrast, black-box attacks do not have access to the model’s parameters and generate a replica of the model by repeatedly querying the input and output. Once the parameters are acquired, they train a substitute model with perturbed data and attack it [39, 90, 117, 137, 145]. Generating perturbations in textual data is more challenging than in images due to the discrete nature of the data [145]. The quality of adversarial examples generated for text data is determined by two factors, namely, the naturalness of the adversarial examples and the efficiency to generate these examples [62]. Some researchers have succeeded in detecting perturbations in text using simpler techniques such as spell-check and adversarial training [89], while others who used word-level attacks failed to efficiently generate adversarial examples due to the high-dimensional search space [138]. Thus, efficiently generating adversarial attacks in NLP poses unique challenges. Despite the difficulty, stronger and imperceptible adversarial attacks have been proposed, which pose a significant threat to the security of deep neural networks [4, 11]. Consequently, several defense mechanisms have been proposed in recent years to counter adversarial attacks in NLP, and the considerable amount of work in adversarial defenses has provided good competition to the novel adversarial attack algorithms, substantially improving the robustness of existing deep learning models.

Adversarial defense strategies in NLP can be broadly classified into three categories: adversarial training-based, perturbation control-based, and certification-based methods. The majority of the work in this field employs an adversarial training approach, and techniques are further subdivided based on the generation of adversarial instances or noise in the defense pipeline. These methods include data augmentation-based adversarial training, adversarial training as a regularization technique, **Generative Adversarial Network (GAN)**-based adversarial training, **Virtual Adversarial Training (VAT)**, and **Human-In-The-Loop (HITL)** approaches. Perturbation

control-based methods are also categorized into perturbation identification and correction and perturbation direction control. Certification-based techniques fall under the third category and provide certificates of robustness against adversarial attacks. A few methods do not fit into any of the aforementioned categories and are classified as miscellaneous. In the subsequent section, the objectives of this survey article are emphasized, distinguishing it from previous surveys.

1.1 Goals of This Survey Article

In this article, we reviewed numerous methods of adversarial defenses in NLP proposed in recent years. The key goals of this survey are as listed below:

- Providing a comprehensive review of adversarial defense schemes in NLP by covering schemes for different NLP tasks and bringing the attention of the community to this emerging area.
- Proposing novel taxonomy for adversarial defense methods in NLP for various tasks.
- Accentuating the importance of defense methods for adversarial attacks and as a regularization scheme in deep neural networks.
- Paving the path for future work in this area by highlighting the open issues.

Numerous survey papers have been published in the past discussing adversarial attacks on deep neural networks in both computer vision and NLP. For example, in Reference [128], the authors conducted a comprehensive survey of adversarial attacks on deep neural networks for images, text, and graphs, proposing a novel taxonomy to categorize a wide range of methods. Similarly, Reference [1] proposed a taxonomy for different adversarial attacks and defenses on various computer vision algorithms, including image classification, image segmentation, object detection, robotic vision, and visual question answering. In addition, [84] presented a brief survey on general adversarial attack methods in deep learning, while Reference [54] briefly reviewed attack and defense methods in images and text data. Furthermore, References [13] and [9] discussed adversarial attacks and defenses for various computer vision algorithms.

In contrast to previous research, Reference [142] proposed a novel taxonomy for universal adversarial attacks, which includes universal perturbations for image classifiers, and briefly discussed attacks on text and audio classification models. While the previously discussed survey papers focused primarily on adversarial attacks on images and briefly discussed attack algorithms in NLP, References [39, 90, 117, 145] extensively reviewed adversarial attack algorithms for various NLP tasks while briefly discussing some defense methods. However, the importance of adversarial defense algorithms is self-evident, given the large amount of work in this area in recent years. Therefore, this survey article aims to address this gap and is different from previous survey papers in this area by focusing exclusively on adversarial defense methods in NLP. This article proposes a detailed and novel taxonomy for adversarial defense mechanisms in NLP, emphasizes the importance of defense methods, and discusses open issues in this area while presenting future work to the community.

In this article, Section 2 discusses adversarial attacks in deep learning and categorizes adversarial attacks in NLP. Section 3 presents a novel taxonomy for adversarial defense methods in NLP. Section 4 provides a detailed description of adversarial training-based defenses in NLP, along with sub-categories. Section 5 discusses perturbation control-based adversarial defense methods. Section 6 outlines certification-based adversarial defenses. Section 7 describes various other adversarial defenses that do not fit into the previous categorization. Section 8 discusses different metrics used to evaluate defense mechanisms. Section 9 describes the datasets and frameworks proposed for training and evaluating adversarial defense methods. Section 10 provides suggestions for future research in adversarial defenses for NLP, and, finally, Section 11 concludes the article.

2 A GENERAL OVERVIEW OF ADVERSARIAL ATTACKS

An adversarial attack is a deliberate attempt to corrupt a deep neural network's functionality by introducing distorted inputs that cause the model to fail. These perturbations are designed to be subtle enough to evade human detection but effective enough to deceive a neural network. For instance, image classification models have been subjected to experiments with various input perturbations, including the addition of noise, the adjustment of pixels, the use of patches, the addition of watermarks, and so on, which can go unnoticed by humans. In contrast, adversarial attacks in NLP involve multiple proposed perturbations at the character, word, or sentence-level through deletion, insertion, swapping, flipping, use of synonyms, concatenation with characters or words, insertion of numeric or alphanumeric characters, and so on. However, it is more challenging to generate adversarial perturbations for text data than image data, because altering a character or word in a sentence is more perceptible to humans. Moreover, creating imperceptible adversarial attacks is difficult in NLP, since perturbations in textual data could result in less natural input data [62]. Adversarial attacks are classified into two categories based on motivation: targeted attacks and non-targeted attacks. Targeted attacks aim to misclassify inputs to a specific class, while non-targeted attacks aim to push the classifier boundary to cause the model to misclassify inputs. Based on access to the model's parameters, adversarial attacks are classified as white-box and black-box attacks. In this section, we briefly review the state-of-the-art adversarial attacks for NLP tasks algorithms.

2.1 Adversarial Attacks in Deep Learning

The goal of an adversarial attack is to generate such perturbations for input $\mathbf{x}$ belonging to class C_1 , such that $\mathbf{x}$ is wrongly classified to class C_2 with a high confidence value. For a multi-class classification algorithm, for k input classes $i = C_1, C_2, \dots, C_k$, the perturbed input is $\mathbf{x}'$ and $\mathbf{f}_i$ is the discriminant function that defines the classification boundaries, where $\mathbf{x}$ belongs to class C_i , and C_{target} is the target class after the attack, then:

$$f_{\text{target}}(\mathbf{x}') > f_i(\mathbf{x}'). \quad (1)$$

Hence, adversarial attacks can be formally defined as an optimization problem for $\mathbf{x}$ as:

$$\begin{aligned} & \underset{\mathbf{x}}{\text{minimize}} (||\mathbf{x}' - \mathbf{x}||) \\ & \text{subject to } \max_{i \neq \text{target}} \{f_i(\mathbf{x}')\} - f_{\text{target}}(\mathbf{x}') \leq 0. \end{aligned} \quad (2)$$

The inequality defined in Equation (1) represents the goal of any adversarial attack that pushes the perturbed input $\mathbf{x}'$ to a desired target class, rather than its actual class. Hence, Equation (2) defines the adversarial attack as an optimization problem, where the goal is to minimize the perturbation magnitude to make the perturbations less perceptible and make sure it gets classified to the target class, C_{target} [76, 84, 92, 128].

2.2 Adversarial Attacks in NLP

In the past few years, numerous methods for adversarial attacks have been introduced, which are specifically designed for NLP tasks. It is important to note that adversarial examples in computer vision cannot be directly applied to text, as they are fundamentally different. Therefore, several attack methods that modify the text data while maintaining imperceptibility to humans have been proposed in the literature. Typically, these methods alter the text data at the word, character, or sentence levels. The following section presents some of these attack methods in NLP.

Character-level adversarial attacks. Character-level attacks perturb the input sequences at a character level. These operations include insertion, deletion, and swapping of characters in a given

input sequence. Despite the fact, these attacks are quite effective, and they can easily be detected with a spell-checker mechanism. One of the techniques used in character-level attacks is adding natural and synthetic noise to the inputs [6]. For natural noise, authors collected natural spelling mistakes and used them to replace words in inputs. For synthetic noise, they swap or randomized characters (except the peripheral) and replace a character with its neighboring character on the keyboard. Adding punctuation marks and increasing or removing the space between characters is another technique to add synthetic noise in the text inputs. For example, in DeepWordBug [28], in black-box setting, they use a two-step process, as they do not have access to the gradients, parameters, or structure of the model. The first step involves finding the most important words in the sentence that would be the target words to perturb. In the second stage, perturbations are added to these select words by the above-mentioned operations. Edit distance is further used to keep track of the readability of the generated sentences. Another example proposed is TextBugger [65] in both black-box and white-box settings where the white-box attack is a two-step process. The first process involves finding the most important word with the help of the Jacobian matrix ($\mathcal{J}_{\mathcal{F}}(x)$) defined as $\mathcal{J}_{\mathcal{F}}(x) = \frac{\partial \mathcal{F}(x)}{\partial x} = \left[\frac{\partial \mathcal{F}_i(x)}{\partial x_i} \right]_{i \in 1 \dots N, j \in 1 \dots K}$, where x_i is the i th word of the input text, N is the total number of words in the input text, and $\mathcal{F}$ is the classifier, and later use five different options to add bugs. These five include insert, delete, swap, substitution with visually similar words, and substitution with a semantically similar words. In the black-box setting, they propose a three-step process where first the most important sentence is identified, then they find the important words to generate five bugs and select the optimal from that. The best adversary is chosen based on how optimal they are for reducing accuracy. Along the same line, Reference [36] has shown that just by adding extra “.” (period), spaces between words, “Perspective” API created by Google gave lesser toxicity scores for the words perturbed in this fashion.

Word-level adversarial attacks. Word-level attacks perturb the whole word instead of a few characters. Common operations include insertion, deletion, and replacement. Word-level attacks can be classified into Gradient-based and Importance-based and replacement-based strategies on the basis of the perturbation schemes used:

- In gradient-based methods, the gradient is monitored for every input perturbation. Whenever the probability of classification is reversed, that particular perturbation is chosen. This is inspired by the **Fast Gradient Sign Method (FGSM)** [31] used for adversarial attacks in computer vision models. If the classification probability changes the class, then the perturbation is considered effective. Another way of using gradient-based method is to find the important words using FGSM and then employ insertion, deletion, and replacement strategies on top of them [100]. Reference [71] used a similar approach where they created adversaries by backpropagating for the cost gradients.
- In importance-based methods it is believed that words with the highest or lowest attention scores play an important role in predictions of self-attention models. Hence, these are chosen as the possible vulnerable words. These words are greedily perturbed until the attack is successful. One of the methods, “Textfooler” [46], uses a similar strategy where important words are greedily replaced with synonyms until the classification label changes. Another work in this direction [40] proposed TextExplanationFooler algorithm, which designed word importance-based attacks for explanation models in text classification problems. Working in a black-box attack setting proposed attack attempted to alter outputs of widely used explanation methods while not changing the predictions of the classifier.
- In replacement-based methods, words are randomly replaced with semantically and syntactically similar words. Here, the replacement for words is obtained by using word vectors such as GloVe [86] or thought vectors. Reference [55] used thought vectors to map sentences to

vectors and replaced one word from its nearest neighbors that had best effect on the objective function. Reference [2] used GloVe vectors to randomly replace words that fit in context of sentence.

Sentence-level adversarial attacks. These attacks can be considered as manipulation of a group of words together instead of individual words in the sentence. Moreover, these attacks are more flexible, as a perturbed sentence can be inserted anywhere in the input, as long as it is grammatically correct. These attack strategies are commonly used in tasks such as Natural Language Inferencing, Question-Answering, Neural Machine Translation, Reading Comprehension, text classification. For sentence-level attacks novel techniques such as ADDSENT, ADDANY are introduced in the literature in recent years with variants such as ADDONESENT, ADDCOMMON [43, 121]. Some of the sentence-based attacks are created such that they do not affect the original label of the input and used as a concatenation in the original text. In these cases, the correct behavior of the model is to retain the original output and the attack is successful if the model changes the output/label. In another set of methods, GAN-based sentence-level adversaries are created that are grammatically correct and semantically close to the input text [148]. Another example, “AdvGen” [15], is introduced, which is an example of gradient-based white-box method and used in neural machine translation models. They used greedy search guided with the training loss to create the adversarial examples while retaining semantic meaning. Another work in this direction [41] proposed **syntactically controlled paraphrase networks (SCPNS)** for adversarial example generation where they used encoder-decoder network to generate examples with a particular syntactic structure.

Multi-level adversarial attacks. Multi-level attack schemes consist of a mixture of some of the methods discussed above. These attacks are used to make the inputs more imperceptible to humans and to have a higher success rate. Hence, computationally more intensive and more complicated techniques such as FGSM have been used to create adversarial examples. In one such method, they create hot training phrases and hot sample phrases. In this method, the training phrases focus on what and where to insert, modify, or delete by finding hot sample phrases in white- and black-box settings where deviation score is used to find the importance of the words [72]. Another example used “HotFlip” [24], which is a character-level white-box attack swapping characters based on the gradient computation. Similar to many other techniques, TextBugger [65] tries to find the most important word to perturb using a Jacobian matrix in a white-box setting. The important words after identification are used for creating adversaries by inserting, deleting, and swapping along with Reinforcement Learning methods with an encoder-decoder framework.

3 TAXONOMY OF ADVERSARIAL DEFENSES

In this section, we will discuss the different types of defense methods used to protect deep learning models from adversarial attacks. We will also highlight some of the recent research works that have shown promise in this area. Adversarial defense strategies are methods used to prevent deep neural networks from failing due to adversarial attacks. These methods aim to increase the robustness of neural networks by training them in an environment that simulates adversarial attacks or by adding mechanisms to detect and handle adversarial inputs. Another approach to increase robustness is to create a perturbation-resistant region around the input space. Therefore, in NLP, there are three main strategies for designing adversarial defense methods: creating a similar environment during neural network training, identifying malicious inputs during training and correcting them using specialized methods, and certifying the robustness of the input region for the network.

The defense methods in NLP discussed in this article are divided into three main categories: adversarial training-based methods, perturbation control-based methods, and certification-based

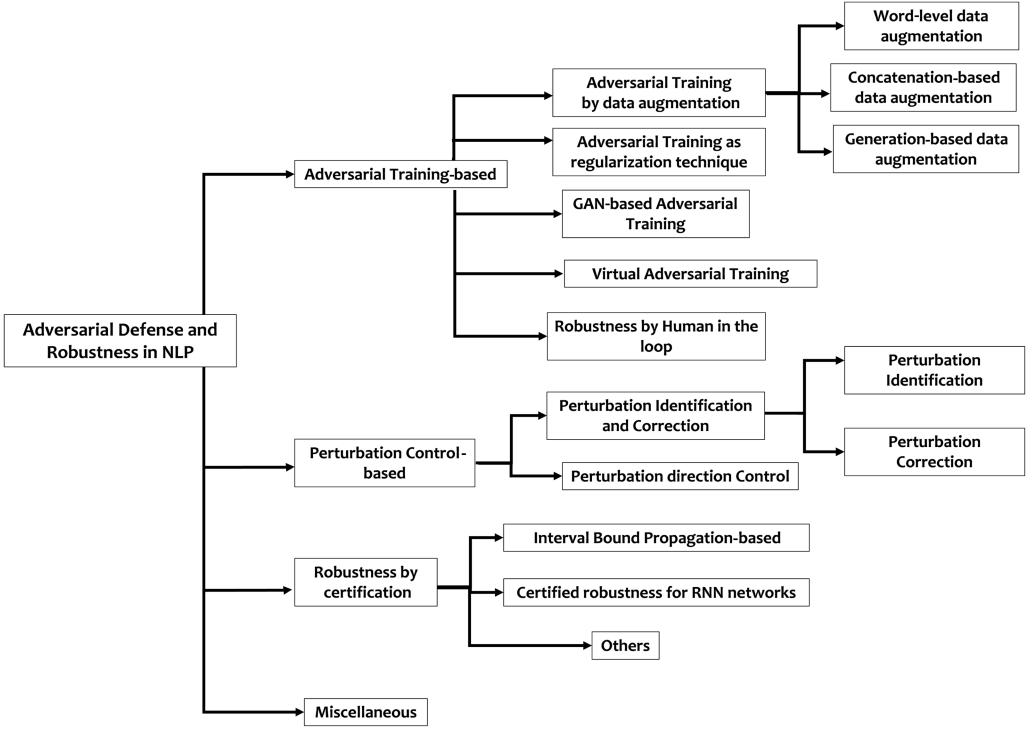


Fig. 1. Taxonomy of adversarial defense methods in natural language processing.

methods, along with some miscellaneous approaches. Methods that do not fall under the first three categories are included in the miscellaneous category. The first set of techniques falls under the category of (i) *Adversarial Training*, which serves as a defense mechanism against adversarial attacks. The various subcategories of methods included in this group are: Adversarial training through data augmentation, Adversarial training as a regularization technique, Adversarial training using **Generative Adversarial Networks (GANs)**, Virtual adversarial training, and Robustness through human intervention. The second set of methods is based on (ii) *perturbation control* and includes two subcategories: Perturbation identification and correction and Perturbation direction control. The third set of methods follows approaches that provide (iii) *certification* of the model's robustness against adversarial attacks. The last set, (iv) *miscellaneous*, comprises a combination of various methods that do not fit into any of the aforementioned categories. Figure 1 illustrates the detailed taxonomy proposed in this survey article for adversarial defense techniques in NLP.

4 ADVERSARIAL TRAINING-BASED DEFENSES

Adversarial training was first introduced in the work proposed in Reference [30]. It is a method of defending against adversarial attacks by introducing adversarial examples in the training data. The strength of adversarial examples decides the final robustness and generalization achieved by the model. Adversarial training is further divided into sub-groups on the basis of the strategies used for augmenting the data such as word- or character-level modification, model-based generation of adversarial examples, or adversarial inputs generated by concatenation in the original data. While some of the methods performed adversarial training by generating a set of adversarial examples and inserting them in the training dataset, other methods used adversarial training as regularizer

by introducing perturbation within the network. In this section, some of the work in the literature will be highlighted that proposes to use of data augmentation for generating adversarial examples for adversarial training.

4.1 Adversarial Training by Data Augmentation

Adversarial defense methods often use adversarial training as a basic technique for defending against adversarial attacks. This method involves generating a set of adversarial data using a perturbation scheme and incorporating it into the model training process. Many techniques for generating adversarial examples using data augmentation have been proposed in the literature. These techniques involve identifying the most important words, characters, or other parts of the input text that affect the output the most and manipulating the input data by flipping, inserting, deleting, or swapping those parts of the sentence. Concatenating a piece of text in the input by finding the most appropriate position is another strategy used in the literature. There are also automatic methods for generating adversarial examples for adversarial training, some of which will be discussed in the following sections.

4.1.1 Word-level Data Augmentation. There are methods in the literature that propose to modify words or word embeddings in the input text for generating adversarial examples to augment data. In this line, the work proposed in TextBugger [65] presents adversarial training by data augmentation for text classification in both white-box and black-box settings by generating utility-preserving adversarial examples. In the white box, important words for perturbations are found using Jacobian of classifier and then optimal perturbations are found by searching the nearest neighbor space in Word2Vec embeddings. In black-box setting, data is augmented by finding important words and sentences that contribute to the output the most and manipulating them. The work in Reference [16] proposed novel data augmentation technique for adversarial training in machine translation task, which reinforces the model to virtual data points around the observed examples in training data. They proposed vicinity distribution for adversarial space (space of adversarial examples centered around each training example) and sampled virtual adversarial samples from it using interpolated embeddings of existing training samples. In Reference [68], adversarial perturbations are applied on word embedding layer of a CNN for text classification task to make the classification model robust towards the worst perturbations. Another work [133] proposed a new method, PQAT, that perturbs the embedding matrix rather than the word vector for machine reading comprehension task. Two additional independent embedding spaces for **paragraph-question (PQ)** are used to give additional context for the same word used with different roles in paragraphs and questions. During training, P-Q embeddings are added to the original vector, keeping the context from passage and question separate.

In another work [144], the authors proposed **continuous bag-of-word (CBOW)** embedding-based perturbations for generating human-imperceptible adversaries for text classification task. Embedding space generated by CBOW is used for predicting the perturbation direction and tries to preserve the meaning of the sentence by placing a constraint on the perturbation direction. Authors generated adversarial examples without altering the semantic meaning of the sentence and used these examples in adversarial training. In the same line, Reference [73] proposed a solution to the **Out-Of-Vocabulary (OOV)** words problem faced by conventional character-level defense methods leading to a poor performance of models. They proposed adversarial stability training to overcome these challenges. Stability training is a technique that makes the output of the neural networks significantly robust while maintaining the original performance [149]. Proposed **Adversarial Stability Training (AST)** is used with character-level embeddings to overcome OOV problems, and adversarial sentences are generated by perturbing each word using character-level

embeddings representation to overcome the distribution problem. Another work in the same direction [37] created adversarial examples using several schemes such as random word replacement, synonym replacement, finding weak spots in the input strings with greedy approach, by constraining the embeddings within L_1 distance, replacing the word on the basis of attention score (high attention score word to low score word) and demonstrate their results on sentiment analysis, textual entailment, and machine translation tasks. They analyzed the robustness of RNNs, transformers, and BERT-based models and demonstrated that self-attentive models are more robust than RNNs. Work proposed in Reference [7] evaluated adversarial training with adversarial examples for eight datasets in NLP targeted for several purposes, such as question answering, reasoning, detection, sentiment analysis, and language detection using language models such as LSTM and GRU. They used combinations of dropout and adversarial example for evaluation. Another work [105] proposed data augmentation for adversarial training for increasing the robustness of causal reasoning task. They proposed data augmentation by synonym substitution and by filtering out casually linked clauses in the larger dataset and used generative language models to generate distractor sentences as potential adversarial examples. To improve the conventional adversarial training methods, Reference [118] proposed to use gradient-based approach for ranking the important words in the training dataset and distilBERT similarity score for finding similarity between two-word embeddings for a faster and low-resource requirement-based adversarial training. Also, they propose to use a percentage of training data for generating adversarial examples instead of converting all the training data for cheaper adversarial training.

In Reference [23], authors have proposed a white box-based defense method by generating adversarial examples with flip, swap, insert, and delete at character level and used gradient-based optimization method to rank the examples. Reference [143] proposed **Metropolis-Hastings (MH)** sampling [17]-based adversarial example generator for text classification. Three level word operations, replacement, insertion, and deletion, are performed with MH sampled words in a black-box setting, while the gradient of the loss function is inserted in the pre-selection function of MH in case of white-box attack. For the text classification task, in Reference [21], adversarial training is used for cross-lingual text classification and robustness enhancement by training it on English data and using the model to predict on non-English unlabelled data. The predicted outputs are used as adversarial examples for adversarial training. In the same line [93], authors have proposed a greedy algorithm probability weighted word saliency for adversary generation for the text classification task. Adversarial examples were generated with word synonym replacement and named entities with other named entities using WordNet, picking up the words that cause maximum change in text classification probability. In Reference [138], black-box adversary generation is proposed, which uses sememe (minimum semantic unit in linguistics)-based word substitution, that is, more sememes per sense mean fewer substitute words that share the same sememes can be found, which negatively affects the adversarial attack success rate. Later, they used Particle swarm optimization [51]-based algorithm to search the optimal adversarial example. In literature, frameworks and APIs have also been proposed, which provides a complete platform to the user with various kinds of attacks to generate adversarial examples to be used for adversarial training for defense. In this line, Reference [80] presented a Python framework for attack generation and defending the model, which augments the data using a word embedding, word swap, thesaurus word swap, homograph character substitution, and so on. They also provide a set of constraints on the generated perturbations to keep them indistinguishable from the original such as, grammar check, POS tags, and so on. Along the same lines, Reference [110] generated imitation models for Google API of machine translation and generated adversarial samples using techniques such as flipping, replacing malicious nonsense, substituting phrases that cause incorrect translation, and attacking original model in a black-box manner. Later, they used those adversarial examples to train the imitation

model and transfer the examples to the victim model. In this work, the authors aim towards finding vulnerability in the victim model to make it more robust by having the victim model output a different high-accuracy translation. Apart from word-level perturbations, there are some sentence-level perturbations proposed to generate adversarial examples, which are discussed in the coming section. These adversarial examples are hand-crafted and generated by identifying vulnerability of model, and further used with adversarial training to defend model from such attacks. However, some papers that proposed model-based generation of adversarial examples are also discussed in coming sections.

4.1.2 Concatenation-based Data Augmentation. Another approach to generating adversarial examples for data augmentation involves using concatenation-based strategies and automatic generation of adversaries with language models. For instance, Reference [44] developed concatenative adversarial perturbations, AddSent, to generate adversarial examples for reading comprehension systems. This method involves concatenating grammatically correct sentences to end of a paragraph, which look like questions or other arbitrary sentences. They also replaced some answers with fake answers that have the same part of speech type and category. Similarly, Reference [122] proposed AddSentDiverse, which generates adversarial examples with significantly higher variance by varying placement of perturbations. They also expanded set of fake answers used in AddSent and demonstrated the limitations of the original AddSent method.

4.1.3 Generation-based Data Augmentation. Another approach to improving the robustness of machine learning models is through generation-based adversarial examples. For example, Reference [48] proposed using knowledge-guided rules and a seq2seq model to generate new hypotheses from a given premise for textual entailment models. In a similar vein, Reference [34] proposed a defense strategy using adversarial training, using a seq2seq model to generate perturbations for structure prediction tasks that involved predicting POS tags, parse-trees, and noun phrases. In Reference [130], the authors proposed a grey-box adversarial attack for sentiment analysis that uses a generator model for data augmentation. Adversarial training was conducted by augmenting data using a static copy mask mechanism in the generator, and counter-fitted word embeddings and label smoothing methods were used to better capture lexical relations and preserve the labels of adversaries. Reference [81] introduced the idea of injecting antonymy and synonymy constraints into vector space representations. In the same line, Reference [114] proposed a new method of adversarial example generation by controlled adversarial text generation where they aimed to perturb input for a given task by changing other controllable attributes of the dataset. For example, in the case of sentiment analysis task for product reviews, product category becomes a controllable attribute that cannot change the sentiment of a review. Their pretraining module consists of an encoder-decoder architecture, which is used to teach the model to copy the input sentence S , assuming that it has the controllable attribute (a) in the sentence. The decoder module is updated to generate a sentence containing attribute $a' \neq a$. In the optimizing module, the subspace of all a' is checked by computing the cross-entropy loss to find the highest perturbation. In similar lines of developing robust NLP model, the work proposed in Reference [56] studied a group of linguistic rules to demonstrate local semantic robustness within a sentence and generated variations in input text using predefined template with fixed label. These templates adhere to the linguistic rules discussed in the paper, where incoming variations will not be able to change the output label and further used them for training robust sentiment analysis models. In the direction of paraphrasing-based adversarial instance generation, Reference [42] proposed **syntactically controlled paraphrase networks (SCPNs)**, which generate a paraphrase of the given sentence with the desired syntax in a controlled manner. SCPNs use a bidirectional LSTM and a two-layer LSTM augmented with soft attention over the encoded states-based encoder-decoder architecture, utilizing a paraphrase pair

and a target syntax tree as the inputs. Using the adversarial instances in adversarial training, they evaluated the proposed method on sentiment classification and textual entailment applications.

Table 1 shows the summary of all the adversarial training methods with data augmentation. It summarizes the strategies used in the literature for perturbation generation, along with the granularity of the perturbations used, demonstrated NLP applications, and the kind of the threat model they are defending. Many of the word-based methods for data augmentation involve operations such as flipping, swapping, inserting, deleting, and synonym substitution with words to modify the original inputs and generate adversarial examples for use with adversarial training. Some methods use these operations with characters in the sentences instead of whole words. These adversarial perturbation generation methods aim to maintain the naturalness of the input text so the perturbation is imperceptible to humans. Therefore, some of these methods use word embeddings for perturbations instead of words from the input text. Another set of methods in the literature use concatenation operations within input sentences by identifying the appropriate position for concatenation, and these adversarial examples are used in adversarial training. In addition to manual perturbations and adversarial example generation, another direction of methods uses generative models to create adversarial examples or perturbations. Along with the data augmentation methods, adversarial training methods introduce perturbations within the training loss functions, which are discussed in the next section.

4.2 Adversarial Training as Regularization Technique

In this section, another defense method based on adversarial training for NLP is discussed. In this style of adversarial training, the input perturbations are incorporated as a part of model training, instead of training it with adversarial examples. In Reference [30], authors proposed adding perturbations in input as a regularizer in the loss function. The modified optimization function based on the fast gradient sign method after adding perturbations is defined as:

$$L_{adv}(x_l, \theta) = D[q(y|x_l), p(y|x_l + r_{adv}, \theta)],$$

$$r_{adv} = \underset{r: \|r\|_2 \leq \epsilon}{\operatorname{argmax}} D[q(y|x_l), p(y|x_l + r, \theta)],$$

where L_{adv} is the adversarial loss term, r_{adv} is the adversarial perturbation, x_l is the labeled input data, D is the non-negative divergence measurement function between two probability distributions, ϵ is the upper bound on the perturbations, $q(y|x_l)$ is the unknown true distribution of the output label. This loss function is designed to approximate the true distribution $q(y|x_l)$ by a parametric model $p(y|x_l, \theta)$ that is robust against adversarial attack to the input data x . Adversarial training using perturbations in the loss function has been shown to be a successful defense strategy, as it generates adversarial examples that are difficult to create manually by exploiting flaws in the optimization function to improve model generalization. In the field of NLP, several techniques have been proposed to introduce perturbations during training or modify the loss function. Here, we describe some of the methods that have been proposed to introduce perturbations during training.

In Reference [78], the authors included perturbations at each step of training and tried to minimize the loss function for text classification tasks. In another work [134], authors proposed adversarial training for POS tagging by using character-level embeddings with BiLSTM models, where word-level embeddings are generated by concatenating character-level embeddings. Perturbations are added to the input at the character-level embeddings in the direction that maximizes the classifier loss function while training the model. Another method [126] introduced adversarial noise at embedding (concatenation of word and characters) level for the task of relation extraction within the multi-instance multi-label learning framework. They proposed a joint model to entity recognition and relation extraction using adversarial training as a regularization scheme where

Table 1. Summary of the Adversarial Training Methods by Data Augmentation

Strategy	Work	Granularity	Application	Threat Model
Word	[65]	Words & Sentences	Sentiment analysis	White Box, Black Box
	[16]	Sentence level	Machine translation	Black Box
	[68]	Words	Text classification	White Box
	[133]	Embedding matrix	Reading comprehension	White Box
	[81]	Word embedding	Dialogue state tracking	White Box
	[144]	Word embeddings	Text classification	White Box
	[73]	Character embeddings	Text classification	White Box
	[37]	Word	Sentiment analysis, Textual entailment, Machine translation	White Box
	[7]	Word and Character embeddings	Question answering, Reasoning, Sentiment analysis, Language detection	White Box
	[105]	Word level	Causal relation classification	White Box
	[23]	Character level	Machine translation	White Box
	[143]	Word level	Text classification	Black box
	[93]	Word level	Text classification	White Box
	[138]	Word level	Text classification	Black Box
	[110]	Phrase level	Machine translation	Black Box
	[21]	Word level	Document & Intent classification	White Box
Concatenation	[44]	Sentence	Reading comprehension	White Box
	[122]	Sentence	Reading comprehension	White Box
Generative	[48]	Sentence generation	Textual entailment	White Box
	[34]	Sentence generation	Predicting POS tags, parse trees, NP	Black Box
	[130]	Sentence generation	Sentiment analysis	Grey Box
	[114]	Sentence generation	Sentiment analysis	White Box
	[56]	Sentence generation	Sentiment analysis	White Box
	[42]	Sentence generation	Sentiment classification, Textual entailment	White Box

worst case perturbations are added to maximize the training loss. Along the same line [113], authors improved the neural language modeling by using adversarial training as a regularization technique. They injected an adversarial perturbation on the word embedding vectors in the softmax layer of the language models while training the model. Authors have suggested new loss functions and diverse neural network optimization methods to train a model adversarially and improve its robustness. In this direction, Reference [77] proposed adversarial training for natural language inferencing by reducing the adversarial example generation problem to combinatorial optimization problem. They proposed a continuous inconsistency loss function that measures the degree to which a set of examples can cause a model to fail. By maximizing the inconsistency loss and constraining the perplexity of the generated sentences, adversarial examples are generated, posing it as an optimization problem. In the same direction, Reference [49] proposed defense mechanism, diversity training, for transfer-based attacks for the ensemble of models. They proposed **gradient alignment loss (GAL)**, which is used as regularizer to train an ensemble of diverse models with misaligned loss gradients. Another work [20] proposed a novel **Adversarial Sparse Convex Combination (ASSC)** method to leverage regularization term for introducing

Table 2. Summary of Adversarial Training as Regularization Technique

Work	NLP task	Granularity
[78]	Text classification	Word embedding
[134]	POS tagging	Character embeddings
[126]	Relation extraction	Word and character embeddings
[113]	Machine translation, Language modeling	Word embeddings
[77]	Natural Language Inferencing	Sentence embedding
[20]	Sentiment analysis and Natural Language Inferencing	Word embeddings
[112]	Question answering, Natural Language Inferencing	BERT embeddings
[153]	Natural Language Inferencing, Textual entailment	Word embeddings
[70]	Sentiment analysis, Text classification	Word embeddings

perturbations. They modeled the word substitution attack space as a convex hull of word vectors and further proposed ASSC-defense for using these perturbations in adversarial training.

There are methods in the literature that proposed novel regularization techniques for enhancing the robustness of a language model. In this line of work, Reference [112] proposed InfoBERT for increasing the robustness of BERT-based models by analyzing language models from information-theoretic perspective. They presented two mutual information-based adversarial regularizers for adversarial training. Information Bottleneck regularizer which extracts minimal features for downstream tasks and removes noisy and vulnerable information for potential adversarial attacks. Other one is Anchored Feature regularizer extracts strong local features that are not vulnerable while aligning local features with global features to increase the robustness. Another work [153] proposed FreeLB, to use K-step PGD [3] for generating adversaries in adversarial training and used multiple PGD iterations. The free adversarial training [102] algorithm eliminates the overhead cost of generating adversarial instances by reusing the gradient information computed when updating model parameters. In this method, while optimizing the objective function, it replaces the batch of input with K times large batch that include perturbations along with inputs. Improving the performance of this work, Reference [70] proposed FreeLB++ by extending the search region to a larger l_2 -norm and increasing the number of search steps at the same time, since FreeLB has a narrow search space. They also bench-marked various defense methods in the literature under standard constraints and settings to have a fair comparison of these methods. Table 2 describes the summary of adversarial training as regularization technique. In the coming section, GAN-based adversarial defense methods are discussed, where GAN is used as a generator and discriminator model to build more robust models against adversarial attacks.

4.3 GAN-based Adversarial Training

Using GAN for adversarial training is another approach that has been used in the literature for defending models from adversarial attacks. In this approach, a generator and discriminator are trained together in adversarial manner, where the generator is primarily used for generating adversarial examples. The discriminator is responsible for discriminating between the clean data and the adversarial training samples to make model more robust towards adversarial attacks. In this line, the work proposed in Reference [48] uses adversarial training with GANs for the textual entailment task. The generator and discriminator are trained in an end-to-end manner, where generator (seq2seq) is trained for generating adversarial examples using external knowledge or handwritten rules. Discriminator is trained in the same manner to learn the textual entailment for the generated samples. In another work [94], authors used a conditional variational autoencoder that generates the adversarial examples for text classification task. They further

Table 3. Summary of GAN-based Adversarial Training Methods

Work	NLP Task	Strategy
[48]	Textual entailment	Using external knowledge
[94]	Text classification	Using Conditional Variational Autoencoder (VAE)
[74]	Text classification	Adversarial shared-private model
[129]	Sentiment analysis	Using lexical knowledge base
[18]	Sentiment analysis and Topic classification	Privacy measurement of neural representations

used a discriminator and GAN training framework for adversarial training and to make sure the generated adversaries are consistent with the real-world data. In Reference [74], authors used multi-task learning for domain adaptation. They proposed a model that has separate units to discriminate between shared patterns and task-specific patterns using GAN for creating adversarial examples and includes this loss in the final loss for optimization. They demonstrated the improved performance over 16 datasets and the learned parameters in both shared and task-specific parts of the network. The work in Reference [129] proposed a new adversarial training approach that mimics a GAN. The generator is used to create adversarial examples with the help of lexical knowledge base where a classifier is used to score the generated adversarial example. The model is trained in a reinforcement learning fashion due to discrete-time generation of the generator model, and the score from the classifier is used as the reward for the generator. The generator generates examples by replacing words in the input sentence with synonyms, neighboring words, and superior words. In another work, [18] try to defend against an adversarial attack which tries to take the encoded information to reconstruct the original input text. In the adversarial training-based defense strategy, they used GAN-style training with two components in which the original one predicts the class of a given sentence and a binary classifier predicts the privacy element. The training style involves making the output prediction as correct as possible while at the same time creating more complex examples for the privacy element classifier. Table 3 describes the summary of GAN-based adversarial training methods, where each row describes the NLP task for which defense is designed and the specific strategy used in the proposed work.

4.4 Virtual Adversarial Training (VAT)

Virtual Adversarial Training (VAT) is another variant of adversarial training-based defense methods first proposed by Reference [79]. VAT is found to be a very efficient method in the case of semi-supervised learning methods, because it defines the adversarial direction without label information. In contrast to Adversarial training, VAT does not need full label information for generating perturbation. The intuition behind VAT is to add perturbation ' r ' to input x such that the divergence of their output space is maximum. Hence, training is done in a way to minimize the divergence term after adding the perturbed input to make the model robust against adversarial attacks. The modified loss function for virtual adversarial training is defined as:

$$\begin{aligned}
 LDS(x_*, \theta) &= D[p(y|x_*, \hat{\theta}), p(y|x_* + r_{adv}, \theta)] \\
 r_{adv} &= \underset{r; ||r||_2 \leq \epsilon}{\operatorname{argmax}} D[p(y|x_*, \hat{\theta}), p(y|x_* + r_{adv})] \\
 R_{adv}(D_l, D_{ul}, \theta) &= \frac{1}{N_l + N_{ul}} \sum_{x_* \in D_l, D_{ul}} LDS(x_*, \theta),
 \end{aligned}$$

where x_* are the “virtual” labels that are probabilistically generated and r_{adv} are virtual adversarial perturbations. Here, x_* is kept in the place of x , since label information is not available for all the

Table 4. Summary of Virtual Adversarial Training (VAT)-based Defense Methods

Work	NLP task	Granularity
[78]	Text classification & Sequence modeling	Word embeddings
[87]	Natural Language Inferencing (NLI)	Word embeddings
[75]	Question answering, NLI, Named Entity Recognition(NER)	BERT embeddings
[144]	Sentiment classification	Word embeddings
[14]	Chunking, NER, slot filling	Character and word embeddings
[67]	NER, NLI, Textual entailment, Text classification	Token level

input data. LDS is the local smoothness term for the input data point x and R_{adv} is the final regularization term.

The work proposed in Reference [78] extended the notion of virtual adversarial training and adversarial training for text classification and sequence models proposing this technique as a regularization method. For defending the models, they introduced perturbations in word embeddings of the text inputs while minimizing the KL divergence of VAT. In another work [87], authors proposed a VAT method by performing adversarial steps on those examples that are predicted as wrong by the model and then regularize the model for this target direction in contrast with general adversarial training methods where the perturbation is done for all examples with variation from the gold label. In a targeted training manner, they try to steer the examples to a particular label y_t and present a comparison with human-annotated data along with other adversarial training algorithms. In the same direction, Reference [75] proposed a novel adversarial robust model **Adversarial training for large neural LangUAge Models(ALUM)** for defending BERT-based pretraining language models. It is a general model for adversarial training in pretraining and fine-tuning that regularizes the training objective by applying perturbations in the embedding space that maximizes the adversarial loss. The model is regularized using VAT. Experimenting with different word embeddings using VAT, Reference [144] extended the adversarial training regularization for semi-supervised tasks. They used **continuous bag of words (CBOW)** model for generating word embeddings and restricted perturbation directions for creating adversaries. Targeting specifically sequence labelling tasks in NLP, Reference [14] proposed VAT for sequence labelling task combining CRF, making sequence labelling task more robust. They use CNN layer for extracting character and word embeddings, LSTM for sequence encoding, and CRF decoder layer to incorporate the probabilities of label transition. Introducing more variations to VAT, Reference [67] proposed Token aware virtual adversarial training. In contrast with conventional virtual adversarial training, TAVAT generated token aware perturbations instead of random perturbations to avoid unnecessary noise and take important information carried by tokens into consideration. Table 4 describes the summary of **Virtual Adversarial Training (VAT)**-based methods along with the specified NLP task for their design and granularity of the perturbation.

Along with the several variants of adversarial training methods, there are a few schemes that utilize **Human-In-The-Loop (HITL)** framework, where human-level intervention is considered for adversarial training. Some of these methods are discussed in the next section.

4.5 Robustness by Human-in-the-loop

Human-In-The-Loop (HITL) is an idea of leveraging human intervention while training models or defending them against adversarial attacks. The scheme is extensively used in various fields of artificial intelligence. It takes advantage of both human and machine intelligence for labeling the data and training and validation of models. While it is proven to be an efficient scheme in other areas of artificial intelligence, several authors have tried to use HITL for developing algorithms for adversarial defenses.

Table 5. Summary of the Human-In-The-Loop (HITL)-based Adversarial Training Methods

Work	NLP Tasks	Human interaction granularity
[138]	Sentiment classification, NLI	Manual selection of word substitution-based perturbations
[82]	NLI	Crowd-sourced and tested with models
[124]	NLI	Crowd-sourced and tested with models
[19]	Offensive language detection	BERT with crowd workers
[109]	Question answering	Human annotators and verification

Reference [138] proposed a sememe-based word substitution method to generate perturbations. Apart from using particle swarm-based optimization algorithm to search perturbation for data augmentation, they manually selected 692 valid adversarial samples for adversarial training to further boost the performance. Also, authors in References [82, 124], created a dataset of adversarial examples, ANLI for natural language inferencing task by crowd-sourcing. Adversarial examples are written and verified by human annotators in three stages in loop while getting them tested from high-performing NLI models. They also presented error analysis and discuss annotation scheme and data collection process of ANLI. In Reference [19], authors built a model for offensive language detection in dialogues using human and models in loop. They trained BERT-based model on Wikipedia Toxic Comments dataset and asked crowd workers for marking the messages as offensive if they are wrongly marked safe by the system. This process is performed in multiple iterations to build a robust system. Work in Reference [109] presented a defense method using HITL by proposing human-computer hybrid approach for evaluating the models. They presented a human verification of the question-answering system, where human annotators authored adversarial examples of questions, targeted to fail a model-based Question-Answering system where the generated questions are still answerable by humans. This process is targeted towards building a robust question-answering system by inserting human-authored adversarial examples. Table 5 depicts summary of the HITL-based adversarial training methods in NLP defense. Each row in the table depicts the NLP application for which the defense method is designed and granularity at which human interaction is proposed. Adversarial training-based defense methods are evidently the most popular way to build robust models against adversarial attacks in NLP. It requires the generation of adversarial examples using several techniques and using them for training the model. However, in contrast with training with adversarial examples, there are methods in the literature that are proposed for the detection of perturbations, their correction, and controlling directions of those perturbed instances. Some of these methods are discussed in the next section.

5 PERTURBATION CONTROL-BASED DEFENSES

The adversarial defense methods proposed in previous sections use data augmentation schemes, perturbation generation within training for supervised and semi-supervised tasks, adversaries monitored by human and models in loop, defending the model in a generating and discriminating manner. However, all these schemes do not incorporate the idea of interpretable perturbations or reconstruction of generated perturbations. In the literature, there are schemes that control direction of perturbations to make the perturbations more meaningful, indistinguishable, and re-constructive and further use them in training. Also, another set of methods try to identify the

perturbed inputs and correct them to make the models more robust. In this line, the following sections describe methods that have been proposed in this direction.

5.1 Perturbation Identification and Correction

In literature perturbation, control-based method tries to identify malicious inputs after an attack and correct them. Some of the methods only identify these inputs and filter the training data accordingly, while the other category of methods also corrects these inputs after their identification using methods such as spell-checker or rule-based. Some of these methods are described below.

5.1.1 Perturbation Identification. Perturbations related to word modifications that included insertion, deletion, substitution or swapping of words are identified in several ways. One of those methods is proposed in Reference [119], where the authors proposed defense mechanism, against synonym substitution, calling it **Synonym Encoding Method (SEM)**. They essentially clustered all the synonyms in embedding space with their Euclidean distances and then encoder is layered before input to train the model. Encoder is responsible for identifying all the synonym substitution-based attacks in the model and maps all the synonyms to a unique encoding without adding extra data for training. Improving on this work, Reference [132] proposed a robust adversarial training method called **Fast Triplet Metric Learning (FTML)**. This method tries to cluster the similar embedding and pushes dissimilar embedding where SEM works directly with input texts and establishes no relation with non-synonym clusters. FTML forces each word with a similar meaning to have the same representation in the feature space and pushes away the word with a different meaning. The method incorporates a word-level triplet loss that tries to minimize the distance between a word with its corresponding group of synonyms and maximizes the distance with its non-synonym group. In another work in this direction, Reference [151] proposed **Dirichlet Neighborhood Ensemble (DNE)**, a randomized smoothing method for training a robust model to defend substitution-based attacks. DNE forms virtual sentences by sampling embedding vectors for each word in an input sentence from a convex hull spanned by the word and its synonyms, and it augments them with the training data (mixing the embedding of the original word in the input sentence with its synonyms). In the same line, Reference [5] introduced frequency-aware randomization framework **Anomaly Detection with Frequency Aware Randomization (ADFAR)** for defense against adversarial word substitution. They add an extra module to detect perturbation in the sentences and apply ADFAR only on sentences that are identified as adversarial. This module is added to the language model and uses a multi-task learning procedure. They demonstrated that ADFAR works better than other defenses on four datasets—MR, SST-2, IMDB, MNLI. Reference [120] proposed an adversarial defense scheme by perturbation detection for synonym substitution attacks. They proposed a novel method, namely, **Randomized Substitution and Vote (RS&V)**. The proposed method calls an input text “adversarial example” by randomly substituting some of its words by their synonyms and checking the consistency of the highest voted label for all perturbed examples. If the label is found to be inconsistent with the original label, then it is considered as an adversarial input. Working in a similar direction, Reference [107] proposed to use of random perturbations to defend sentiment analysis models. They inserted random perturbations to the multiple copies of a randomly selected sentence from the reviews. These perturbations included synonym substitution, dropping of a word or a spell-check with correction if necessary. All these perturbed copies of the sentences are put together with the original review and a majority vote is taken for each sentence by the sentiment classifier. By majority-based classification, the models are taken back to their original performance accuracy before the attack. In the direction of perturbation identification, Reference [127] proposed an extensive dataset, TCAB, for attack detection and labeling with over 1.5 million instances of adversaries. To construct this dataset, authors used six datasets

for text classification with three target classifiers and 12 different attacks from textAttack and openAttack and presented a benchmark for these attacks. They proposed an attack identifier/labeler using three features of the input such as text properties, including contextual embeddings, length of the input, token case, punctuation, non-ASCII characters. Authors use language model's properties as another feature that identifies the structure of the language such as phrasing or ungrammatical input text. They used target model's properties as another feature that captured the changes in model's gradients, activation, or saliency because of malicious inputs and trained a classifier with them. Utilizing the TCAB dataset and the text, language, and classifier features, the detection of the perturbation and labeling the type of attack is proposed. Another work [116] proposed a novel method called TextFirewall identification of adversarial inputs. They used word importance to quantify the importance of a word in the input sentence to the final classification of the model. The impact of each word in an input sentence is calculated to detect the perturbation by summing the scores of each model, and then the model's output is compared with original ground truth to identify the perturbed input. In a similar direction, Reference [136] proposed an adversarial robustness method by proposing feature density estimation-based perturbation detection. In contrast with the available frequency-based likelihood estimation, they utilized the probability density of sentences. They proposed a method of **Robust density estimation (RDE)**, which fits the probability density estimation model on the features obtained from a pre-trained model like BERT. Dimensionality reduction is applied to the parameters of these features to avoid the curse of dimensionality. In addition, they released a benchmark for word-level adversarial detection using four NLP models with four different datasets for text classification. In addition to the discussed work in perturbation identification, some of the methods in the literature also attempt to correct these perturbed input data after their detection. Some of the perturbation correction methods are discussed in next section.

5.1.2 Perturbation Correction. Adversarial inputs are also required to be corrected after their identification to retain the training data. Some of the methods in this direction of work attempted to identify perturbations related to character-level modifications in the input text. In this direction, Reference [99] proposed a **semi-character-level recurrent neural network (ScRNN)**, which acts as a spell-checker by recognizing words. ScRNN has an architecture similar to standard RNN and takes semi-character vector as input and predicts a correct word at each timestep by applying three types of noises: jumble, delete, and insert. As an extension of the above work, in Reference [89] authors try to combat misspellings by using a word classifier before the actual classifier of a task. They propose ScRNN with backoff to overcome limitations of ScRNN [99] and propose three backoff techniques if the word classifier predicts it as **unknown (UNK)**. As a backing-off step, the word recognizer either passes the UNK word as is, backs off to a neutral word, or backs off to a more general word recognition model trained on a larger, less specific corpus. In Reference [50] authors demonstrated the limitations of spell-checker for perturbation identification and correction. They proposed a method in which context-independent probability distribution is created by segmenting the perturbed sentence using BERT tokens and modified version of Levenshtein distance. For context dependent probability—all the embeddings of context-independent hypothesis are clubbed into a weighted embedding. Now a token is masked and MLM is used to predict the tokens. This process is repeatedly done for the best approximation. Now, these hypotheses are sent to GPT for getting the language modeling score, and the best hypothesis is selected from that. They compared their method against Pyspellchecker, human annotations, and RNN trained for spell-checking. In Reference [26], authors presented backdoor attacks as adversarial attacks during training of the model and proposed attacking methods for NLG model by inserting trigger words in the input sentence. They further proposed defense strategies by detecting hacked inputs and output correct results and preserving the correct input and giving its output.

Reference [150] proposed a novel method for perturbation identification and correction, in which they try to recover the perturbed token based on the context and with the help of small world graphs. First they use BERT model to get the contextualized embedding vector for each token and then pass it to a binary classifier for classification of perturbation. Later, they used a BERT-based context network to be used as the context for predicting the perturbed word. The perturbed word is masked and passed to the BERT to get the embedding of the mask token. Using the embedding vectors and small world graphs, they recovered the affected tokens. In the direction of perturbation identification and correction, in Reference [8], various types of perturbed text are identified and corrected using rule-based methods such as alternating characters defense, which corrects the combined unicode, space separation, and zero-width space separation perturbation in the entire document. Another rule-based defense used is Unicode Canonicalization, which corrects and replaces unicode and tandem character obfuscation perturbations. They further used a continuous bag of words-based embeddings and identified embeddings that are generated by parts of a single word combining random spaced words followed by a process to find similar embedding for words with similar spelling. A skip-gram model is trained with vectors of similar context to ensure the embeddings having similar spelling and context are closer. They evaluated their embeddings with the downstream task of classifying the Facebook posts as engagement bait or otherwise. In a different line, Reference [152] proposed a universal perturbation detection method, TREATED, to defend against various perturbation levels without making any prior assumptions. They utilized several reference models to make different predictions about clean and adversarial examples and block them if found adversarial. They designed the reference models on the basis of their consistency on the clean and adversarial data. In the direction of identifying perturbations for other language text than English, Reference [63] proposed a defense model for text classification for Chinese language. Adversarial perturbations are detected in three steps. **Neural Machine Translation (NMT)** model is used for removing the noise in the input text. The corrected text is converted into multimodal embeddings (semantics, glyph, and phonetics), and the extracted features are given into text classification.

Table 6 demonstrates the summary of the various perturbation identification and correction methods. It shows the type of attack for perturbing the input data, the strategy used for detecting the perturbations, NLP applications on which the proposed method is demonstrated, and whether or not they are attempting to correct the perturbed input after their detection. As can be seen, a large part of perturbation detection and correction method is limited to synonym substitution and misspelling-based adversarial attacks. Also, the demonstration of the proposed defense is largely demonstrated on various types of text classification tasks, including sentiment classification, news category classification, and topic classification. The commonly used techniques for perturbation correction after their detection includes blocking the perturbed data, generating or predicting the clean text, and replacing it with similar correct words. There are methods proposed in the literature that defend the machine learning model by controlling the direction of perturbations in their embedding space. Some of these methods are discussed in the next section.

5.2 Perturbation Direction Control

The proposed work under perturbation direction control alters the direction of the perturbations towards the cleaner text input limiting the adversarial space. Along this line, Reference [101] proposed an interpretable adversarial training method by restricting the direction of adversarial samples. The direction of perturbation is restricted to the words in the existing vocabulary so perturbations could be interpreted even after adversarial training. In Reference [144], authors propose to use CBOW to predict the perturbation direction while trying to preserve the meaning of the sentence by placing a constraint on the perturbation direction. Another work [95] proposed

Table 6. Summary of Defense Schemes Proposed for Perturbation Identification and Correction

Work	Attack	Method	Application	Perturbation Identification	Perturbation Correction
[119]	Synonym Substitution	Maps synonyms to unique encoding	Sentiment, Topic classification	✓	—
[132]	Synonym Substitution	Cluster similar embeddings	Sentiment, Topic classification	✓	—
[151]	Substitution based	Mixing embeddings of words & synonyms	Sentiment, News category classification	✓	—
[5]	Substitution based	frequency aware randomization	Sentiment classification, Natural Language Inference	✓	—
[120]	Synonym substitution	Randomized synonym substitution & vote	Sentiment, Topic, News category classification	✓	—
[107]	Synonym substitution	Random perturbation defense	Sentiment analysis	✓	—
[127]	Attacks from TextAttack [80, 139]	TCAB attack identification dataset & Text, Language, Classifier properties	Sentiment, Abuse/No-abuse classification	✓	—
[116]	Word-level perturbations	Finding word importance	sentiment classification	✓	—
[136]	Word-level perturbations	Feature density estimation	Sentiment, News categories, Topic classification,	✓	—
[99]	Misspellings	ScRNN- Spell checker	—	✓	✓
[89]	Misspellings	ScRNN with Backoff	Sentiment Analysis	✓	✓
[50]	Misspellings, orthographic attacks	Context independent probability distribution	Restoring sentences	✓	✓
[26]	Backdoor attacks	Trigger word manipulation and BERTScore	Machine translation, Dialogue Generation	✓	✓
[150]	Word perturbations	Recover perturbed tokens with small world graphs	Text classification	✓	✓
[8]	Misspellings	Embeddings similar to original words, Rule-based methods	Engagement Bait Classifier	✓	✓
[152]	Synonym substitution, replacement order strategy	Using reference models	Sentiment analysis	✓	✓
[63]	Word-level perturbations	NMT model is used for removing noise	Text classification	✓	✓

Table 7. Summary of Perturbation Direction Control-based Adversarial Defense Methods in NLP

Work	NLP Tasks	Perturbation
[101]	Sentiment classification	Word embeddings
[144]	Sentiment classification	Word embeddings
[95]	Malware detection	Word embeddings for API call command

an adversarial defense mechanism, Sequence Squeezing, aimed to make RNN models and their variants robust against adversarial attacks. The proposed method generates semantic preserving embeddings that are low in the number of features than the original embedding. The squeezed embedding is tested for adversarial attacks in malware detection and added to the training data while diminishing the adversarial space for generating perturbations. Table 7 presents the summary of perturbation direction control-based adversarial defense methods in NLP, where each row shows the NLP applications and the type of perturbation used in the threat model in the associated work. Proposing adversarial defense for text input data with perturbation direction control is a step towards developing more interpretable defense model than conventional methods of adversarial training. The discussed methods in this category demonstrate their defense scheme on various types of text classification tasks, such as sentiment classification, malware classification. Another direction of adversarial defense methods in NLP propose to provide a certified region of robustness while training their machine learning model. Some of these methods are discussed in detail in the coming section.

6 ROBUSTNESS BY CERTIFICATION

The methods discussed in the previous sections for adversarial defenses involved word/character substitution-based adversaries where words are synonyms to make the perturbation look indistinguishable. Other methods tweak words by inserting characters, changing spellings, and deleting/swapping characters. All these adversarial samples are necessary for defending the models but they are not sufficient. An attacker can generate millions of adversarial examples by modifying every word in a sentence. A defense algorithm based on adversarial training requires a sufficient amount of adversarial data to increase the robustness, which still does not cover a lot of unseen cases that are generated by exponential combinations of different words in a text input. Also, perturbation control-based methods require identification of perturbations on the basis of already-seen perturbations with a prior assumption of the type of attack. These methods have limitations in their performance when model is exposed to new adversarial instances. Hence, there is a separate set of adversarial defense methods in the literature that are driven by “certification.” These methods train the model to provide an upper bound on the worst-case loss of perturbations and hence provide a certificate of robustness without exploring the adversarial space.

6.1 Interval Bound Propagation-based Methods

Interval Bound Propagation (IBP) [32] is a bounding technique extensively used in images for training large, robust, and verifiable neural networks. Training the neural networks with IBP technique tries to minimize the upper bound on the maximum difference between the classification boundary and input perturbation region. IBP lets you include the loss term in the training, using which the last layer of the perturbation region can be minimized and kept on one side of the classification boundary. Now, this adversarial region is tighter enough and can be said certified robust.

In this line, Reference [45] proposed certified robust models while providing maximum perturbations in text classification. Authors used interval bound propagation to optimize the upper bound

over perturbations. IBP gives an upper bound over the discrete set of perturbations over word vector space. IBP computes an upper bound on the model's loss when given an adversarially perturbed input. This bound is computed in a modular fashion. Reference [38] introduced a verification and verifiable training of neural networks in NLP. Authors proposed a tighter over-approximation in the form of a "simplex" in embedding space in the input to generate perturbations. To make the network verifiable, they define it as the convex hull of all the original unperturbed inputs as a space of delta perturbation. Using IBP algorithm, they generated robustness bounds (by generating bounds for each layer). Reference [135] proposed structure-free certified robust models that can be applied to any arbitrary model. This method overcomes the limitations of IBP-based method in which they are not applicable to character-level and sub-word-level models. They prepared a perturbation set of words using synonym sets, top-K nearest neighbors under the cosine similarity of GloVe vectors, where K is a hyperparameter that controls the size of the perturbation set. They further generated sentence perturbations using word perturbations and trained a classifier with robust certification. In the context of IBP methods, Reference [111] demonstrates the lack of generalizability of IBP-based methods for novel contextual embeddings and a wider range of NLP tasks. They demonstrated the performance of their method in the sentiment analysis task.

6.2 Certified Robustness for RNN Networks

Despite having a plethora of work in finding a certificate for robustness, there is a lack of applicability in RNN-based network due to their inherent complexity. Hence, in another line of work for robustness by certification, certified robustness for RNN-based networks and self-attentive networks is proposed. In this line, the work Popqorn [53] proposed certified robustness for RNN-based networks such as LSTM, GRUs. The challenge is to find a certificate of robustness in RNN-based networks with their complex feedback architectures, the sequential inputs, and the cross-nonlinearity of the hidden states. Authors used 2D planes to bound the cross-nonlinearity in LSTMs and proposed to find a certificate within a l_p ball (attack distance) if the lower bound on the true label output unit is larger than upper bounds of all other output units. They generated certificate of robustness by writing all the bounds as a function of epsilon and tried to find the optimum value of epsilon using a binary search procedure. In the work Cert-RNN [22], they overcame the limitations of Popqorn [53] by a robust certification framework for RNNs. The method overcame the limitations of Popqorn by keeping the inter-variable correlation and speeding up the non-linearities of RNN for practical uses. Authors created a zonotope [25] around the input perturbations and used that to be passed through a vanilla RNN or LSTM. The properties of the output zonotope can be verified to be certifiably robust. They used zonotope instead of a box to preserve inter-variable correlation, the precision of the network, and achieve a tighter bound. They could achieve tighter bounds and at least 19 times faster framework than Popqorn. In this line, Reference [146] proposed a novel approach, **Abstractive Recursive Certification (ARC)**, for certified robustness in RNN-based networks. Authors defined a set of programmatically perturbed string transformations and constructed a perturbation space using those transformations proposed in Reference [143]. They memorized the hidden states of the strings in the perturbation space that shared a common prefix to reduce the evaluation of LSTM cells while finding an upper bound to the loss and avoiding re-computing of hidden states. Following that, they represent all the perturbation sets as a hyperrectangle and pass the hyperrectangle through the remaining network using IBP technique [32]. Following a similar direction, the work in Reference [97] presents **Polyhedral Robustness Verifier of RNNs (PROVER)**, which represents the perturbations in input data in the form of polyhedral that is passed through an LSTM network to obtain a certifiable verified network for a more general sequential data. Another work in this direction is proposed by Reference [10], where authors proposed DeepT, an abstract transformer-based network certification method.

They attempted to certify larger transformers against synonym replacement-based attacks. In this work, authors propose to use multi-norm zenotopes improving the precision of standard zenotope-based methods, which works well for longer sentences by certifying larger radii of robustness ($\times 28$ of existing methods). In another work, Reference [103] proposed an algorithm for verifying the robustness of transformers with self-attention layers that include challenges such as cross-linearity and cross-positional dependency. They provide a lower bound to a boundary (delta certificate) that will be always greater than 0 (probability of correct class is always higher than incorrect class) within a set of inputs that also include perturbations and tighter than IBP. They achieved this by computing lower/upper bound for each neuron with respect to the input space.

6.3 Other Convex Optimization-based Methods

There are other methods in the literature for finding certified robustness for neural networks that used several convex optimization schemes and randomized smoothing-based schemes. In this line, Reference [106] proposed certified defense method for text classification task. They consider data sanitation defenses, which examine the entire datasets and try to remove poisoning points. They upper bound the worst possible test loss of any attack, which works in an attacker-defender setting at the same time. They generated a certificate of robustness (upper bound) by inserting perturbed data at the time of training, where defender is learning to remove outliers at each iteration. Upper bound fits all possible points that evade outlier removal. In Reference [91], authors proposed certified robustness method based on semi-definite relaxation. They computed an upper bound on the worst-case loss of the neural networks with one hidden layer. The computed certificate of robustness provides an upper bound on the robustness for all kinds of attacks and being differentiable they trained it jointly with the network. Reference [115] provided a certificate of robustness with the idea of differential privacy in the input data. They implemented differential privacy in the textual data by treating a sentence as a database and words as individual records. If a predictive model satisfies a certain threshold (epsilon-DP) for a perturbed input, then its input should be the same as the clean data. Hence, providing a certification of robustness against L-adversary word substitution attacks.

Reference [140] proposed defense algorithm to overcome the limitations of previous methods with an assumption that perturbation generation methods will be known *a priori*. They proposed RanMASK, a certifiably robust defense method against text adversarial attacks based on a new randomized smoothing technique for NLP models. Manually perturbed input text is given to the mask language model. Random masks are generated in the input text to generate a large set of masked copies of the text. A base classifier is then used to classify each of these masked texts, and the final robust classification is made by “majority vote” and trained with BERT and RoBERTa to generate and train with masked inputs. Another work in this direction [57] estimated the **Maximum Safe Radius (MSR)** for a given input text, i.e., minimum distance between the classification boundary and embedding space. They quantified the robustness of neural networks against word replacement that is based on a minimum safe radius. They approximated the upper bound using Monte Carlo tree search and the lower bound by constraint relaxation technique of MSR for CNN and LSTM networks. In Reference [88], the authors also tried to club the concept of fairness and robustness to increase the robustness of a neural network. They demonstrated that a certified robust model can also be used as a bias mitigation system to build trustworthy NLP systems. They integrated a bias mitigation system with state-of-the-art certified robust models to improve the robustness of a model. Table 8 presents the summary of the robustness by certification methods in NLP, where rows are grouped by the certification method used, and each row describes the NLP application, machine learning model used for certification, and type of perturbations

Table 8. Summary of the Certifiable Robustness Methods in NLP

Certification method	Work	NLP Taks	Models	Perturbations
IBP	[45]	NLI, Sentiment classification	Feed Forward network, CNN, Bi-Directional LSTM, Decomposable attention	Word substitution
	[38]	Sentiment & Topic classification	1 layer convolution network	Word substitution and character typos
	[111]	Sentiment classification	CNN	Word embeddings
	[135]	Sentiment & Text classification	BERT	Word Substitution
RNN-based	[53]	Question classification	LSTM, GRU	ϵ bounded L_p ball
	[22]	Sentiment analysis, Toxic comment detection, and Malicious URL detection	RNN, LSTM	ϵ bounded L_p ball
	[146]	Sentiment classification	LSTM	Word substitution
	[97]	Speech classification	RNN, LSTM	ϵ perturbation, dB perturbation
	[10]	Sentiment classification	Transformer networks	L_p noise where $p \in \{1, 2, \infty\}$
	[103]	Sentiment classification	Transformer	ϵ bound perturbations
Other methods	[106]	Sentiment classification	SVM	ϵn poison points
	[115]	Text classification	LSTM	Word substitution
	[140]	Sentiment & Topic classification	BERT & RoBERTa	Word substitution & Character level
	[57]	Sentiment, News, Topic classification	CNN, LSTM	Word substitution
	[88]	Toxicity, Occupation classification	CNN, BERT	Word substitution

generated in the threat model with each associated work. Currently these methods are not proven to be generalized across different types of deep neural networks, and they have been evaluated for small set of NLP tasks and on smaller networks. There are various methods for defending the neural network from adversarial attacks and achieving robustness that are not discussed in these sections and that follow a different line of approach, described in the next section.

7 MISCELLANEOUS

In the previous section, various methods are discussed for adversarial defenses and robustness enhancement. These methods follow the proposed taxonomy and categorization of adversarial defense schemes discussed in Section 3. However, there are several other schemes proposed in recent years that do not fall into any of the categories discussed above. In the direction of enhancing ro-

bustness by bias reduction, Reference [104] tries to remove hypothesis-only bias for NLI datasets by using adversarial classifiers to detect bias in the sentence representation. They demonstrated that the larger the sentence embeddings, harder it is to remove the bias and requires more adversarial classifiers. They tested models with an ensemble of 20 different classifiers where 8 out of 13 datasets performed better than when using only one adversarial classifier and for 3 of them the performance was same when used with adversaries between 1 and 20.

In another line of defending APIs from adversarial attacks, Reference [35] showed that hosted BERT-based APIs are vulnerable to theft, and users can query the API for a dataset and train a BERT model to replicate the API. The replicated model can then be used for adversarial example transfer. They suggested a parameter-based defense strategy by using a temperature parameter in softmax to smooth the output prediction probabilities. They further add perturbation noise with variance sigma to the output probabilities where the larger the variance, the stronger the defense.

In the direction of creating various adversarial examples for adversarial training, Reference [33] proposed variable length adversarial attack in contrast to an existing method that focuses on fixed length. This is achieved by using a special “BLK” token during fine-tuning and then using three atomic operations, addition, deletion, and replacement, to create adversarial examples. They show that this method successfully attacks the models in NLU and NAT tasks and demonstrated its use for creating augmented data for adversarial training. In the same line, the authors in Reference [26] presented backdoor attacks as adversarial attacks during training of the model for NLG models. They proposed post hoc defense against the attacks by using token removal and token substitution on a sentence and corpus level.

Taking VAT in different directions, Reference [154] proposed a novel strategy, **Stackelberg Adversarial Training (SALT)**, which employs a Stackelberg game strategy. There is a leader that optimizes the model and a follower that optimizes the adversary. In this Stackelberg strategy, the leader is advantageous knowing the follower’s strategy, and this information is captured in Stackelberg gradient. They find the equilibrium between the leader and follower using an unrolled optimization approach.

Another work in the direction of robustness enhancement, proposed in Reference [47], introduced **Robust Encodings (RobEn)**, which is a simple framework that guarantees robustness without making any changes to model architecture. The core component of RobEn is an encoding function, which maps sentences to a smaller, discrete space of encodings on a token level. They attempt to cluster all possible adversarial typos into a single cluster using graph-based agglomerative clustering and try to balance between having too many words in a cluster versus a single word in a cluster. In the same line, for languages other than English, Reference [64] proposed AdvGraph to enhance the adversarial robustness of Chinese NLP models with modified embeddings. Due to the inherent complexity in Chinese language, the existing adversarial defense models are difficult to be extended for the Chinese language, hence they propose to capture the similarity in words using graphs. They constructed undirected adversarial graph based on the glyph and phonetic similarity of Chinese characters and learned the representations through graph embeddings to be used with semantic embeddings to be used for other downstream tasks.

There are various metrics used in the literature for evaluating the robustness of the models against adversarial attacks. Evaluating adversarial robustness is equivalent to evaluating the performance of a model in the presence of adversarial attacks, before and after the implementation of defense mechanism. Hence, some of these metrics are standard performance metrics widely used in machine learning literature. Other variants of evaluation metrics are proposed to measure the performance of certifiably robust models. The next section describes these evaluation metrics used in adversarial defense literature.

8 METRICS FOR EVALUATION

There are various evaluation metrics that are extensively used in the literature for evaluating the proposed defense methods. Majorly, these metrics are performance evaluation metrics for the machine learning model that is required to be defended. Adversarial defense methods are evaluated with the performance of the model after the defense method is implemented or runtime evaluation in case of methods that aim towards optimizing a lower/upper bound. Hence, some of these metrics are accuracy, error, loss analysis, measurement of the success of adversarial attacks, similarity with ground truth in the case of language generation, and so on. In this section, some of these commonly used metrics are described in detail.

- **Prediction accuracy (Conventional Accuracy):** Adversarial defense methods for text classification models, such as sentiment classification, Natural Language Inferencing tasks, are evaluated on the basis of the prediction accuracy after implementation of defense algorithm. The prediction accuracy after defense is compared with prediction accuracy after attack, and if there is a surge in accuracy, the defense method is considered to be successful. It is defined as the fraction of test set that is correctly classified [115]. Conventional accuracy is a standard metric that is used to evaluate any deep learning system, and it can be used to evaluate any defense method.

$$\frac{\sum_{t=1}^T \text{CorrecClass}(X_t, L, \epsilon)}{T}$$

Here, $\text{CorrecClass}(X_t, L, \epsilon)$ gives 1, if test sample, X_t is correctly classified for test data T .

- **Loss function analysis:** The negative log-likelihood (loss function) is tested over its rate for adversarial training as regularization and virtual adversarial training-based methods. It can indicate lower error rate and reduced overfitting in adversarial training-based regularization.
- **Error analysis:** Adversarial defense methods are also evaluated on the error rate in the prediction of the model. The error rate is compared with an adversarial attack before and after the implementation of defense schemes. A lesser error rate after defense method entails its successful defense scheme. Error analysis is a standard metric that is used to evaluate any deep learning system for prediction, and it can be used to evaluate any defense method in the literature.
- **Embedding testing:** Embedding test is done for evaluating the embeddings generated for adversarial training. Similarity metrics such as Edit distance, Jaccard similarity coefficient, and semantic similarity metrics are used to evaluate the utility of the adversarial samples generated by finding their similarity with the original input samples.
- **Human Evaluation:** To measure the utility of the adversarial samples for adversarial training, human evaluation is also performed in the literature [29]. Human annotators are asked to judge the adversarial examples in terms of their naturalness by presenting both original and adversarial examples. The good quality adversarial examples are used in adversarial training and further robustness of the deep neural network is evaluated. Adversarial attacks and defense methods are closely associated, hence, the weaker the adversarial attack, the stronger the defense strategy.
- **Attack Success Rate (ASR):** ASR [125] is a measure of success of the adversarial samples created for the potential adversarial attack. This metric is used to measure the effectiveness of the adversarial attack after any of the defense schemes is implemented. The attack success rate is measured before and after defending the model, and a drop in its value will imply a

more robust model. It is defined as:

$$ASR = \frac{N_{successful}}{N_{Total}} * 100,$$

where $N_{successful}$ is the number of adversarial samples that were able to successfully fail the model, and N_{Total} is the total number of adversarial samples generated.

- **BLEU:** Adversarial defense schemes for natural language generation models utilize performance metrics such as BLEU score [85] for the evaluation of their proposed method. BLEU score is measured using n -gram to evaluate the quality of the generated natural language by comparing it with ground truth. It is defined as:

$$p_n = \frac{\sum_{C \in \{Cand\}} \sum_{gram-n \in C} Count_{clip}(gram - n)}{\sum_{C' \in \{Cand\}} \sum_{gram-n' \in C'} Count(gram - n')}$$

$$BP = \begin{cases} 1, & \text{if } c > r \\ e^{\frac{1-r}{c}}, & c \leq r \end{cases}$$

$$BLEU = BP.exp^{\sum_{i=1}^N W_n \log(p_n)},$$

where (p_n) is the n -gram modified precision score, BP is brevity penalty used for longer candidate summaries and for spurious words in it, c is the length of the candidate summary, and r is the length of the reference summary.

- **Number of Queries:** This denotes average number of times attacker queries the model. The higher the average number of queries made by an attacker, the more difficult it is to fail a defense mechanism [70]. It can be used for evaluating any adversarially trained defense model using adversarial instances or by controlling the perturbations.
- **Precision on certified examples:** Precision on certified examples [58] measures the number of correct predictions exclusively on examples that are certified robust for a given prediction robustness threshold. It is defined as:

$$Precision = \frac{\sum_{t=1}^T (isCorrect(X_t) \& robustSize(p_t, \epsilon, \delta, L) \geq Threshold)}{\sum_{t=1}^T robustSize(p_t, \epsilon, \delta, L) \geq Threshold},$$

where $isCorrect(X_t)$ gives a value of 1 if the input sample X_t is correctly classified, and $robustSize$ gives the certified robustness value for the bound L .

- **F1 Score:** This is a metric that combines precision and recalls both to evaluate the overall performance of the model. It is used for comparison of model's performance before and after defense mechanism implementation. F1 score is a standard metric used for evaluating the performance of deep neural networks, and it can be used to evaluate any defense method. It is defined as:

$$F1\ Score = \frac{2 * (Precision * Recall)}{(Precision + Recall)}.$$

- **Certified Radius:** In certification-based adversarial defense methods, robust radius is the largest radius centered around input sample X_t , for which the classifier does not change its value for its corresponding perturbed sample X_t^{adv} . However, calculating robust radius of a deep neural network is an NP-hard problem [141]. Hence, certification-based methods are tested for their certified radius for different norms of perturbations for targeted model on parameters such as minimum radius, average radius, and time taken to obtain it. Certified radius is a lower bound to the robust radius and leads to a guaranteed upper bound of the robust classification error.

- **Certificate Ratio (CR):** This metric is used in certification-based defense schemes. It is the fraction of testing samples that satisfies the certification criteria after prediction [115]. It is defined as:

$$CR = \frac{\sum_{t=1}^T \text{CertifiedCheck}(X_t, L, \epsilon)}{T}.$$

Here, $\text{CertifiedCheck}(X_t, L, \epsilon)$ gives 1 if the fraction of the test data is certified robust.

- **Certified Robustness:** This metric is used in certification-based defense schemes. Certified Robustness [61] for a particular X_t is the maximum value ρ for which it is certified that classifier will return the correct label where X_t^{adv} is its corresponding perturbed sample, such that $\|X_t - X_t^{adv}\| \leq \rho$.
- **Median Certified Robustness:** This metric is also used in certification-based defense schemes. The Median Certified Robustness [61, 140] on a dataset is the median value of the certified robustness across the dataset. It is the maximum value ρ for which the classifier can guarantee robustness for at least 50% of samples in the dataset. In other words, we can certify the classifications of over 50% of samples to be robust to any perturbation within ρ .
- **Certified accuracy:** This is the metric used for evaluating the certifiable robust models. Certified accuracy [58, 115] is the percentage of correct test samples for a certified robust model for the given perturbation. It denotes the fraction of testing set on which a certified model's predictions are both correct and certified robust for a given prediction robustness threshold. It is defined as:

$$\text{Certified Accuracy} = \frac{\sum_{t=1}^T (\text{isCorrect}(X_t) \ \& \ \text{robustSize}(\text{scores}, \epsilon, \delta, L) \geq \text{Threshold})}{T},$$

where $\text{robustSize}(\text{scores}, \epsilon, \delta, L)$ is the certified robustness size for the bound L and $\text{isCorrect}(X_t)$ give a value of 1 if the input sample X_t is correctly classified for test data T .

- **Conditional Accuracy:** This is the metric used for evaluating the certifiable robust models. Conditional Accuracy is proposed by Reference [115], evaluating the classification accuracy of both a clean sample X_t and its corresponding adversarial sample X_t^{adv} within a bound L . It checks when X_t is certified within bound L , whether X_t^{adv} is also classifying correctly. It is defined as:

$$\text{Conditional Accuracy} = \frac{\sum_{t=1}^T (\text{Certified}(X_t, L, \epsilon) \ \& \ \text{corrClass}(X_t^{adv}, L, \epsilon))}{\sum_{t=1}^T (\text{Certified}(X_t, L, \epsilon))},$$

where $\text{Certified}(X_t, L, \epsilon)$ gives 1, when clean input sample X_t is successfully certified, and $\text{corrClass}(X_t^{adv}, L, \epsilon)$ gives 1 when its perturbed input X_t^{adv} is also correctly classified.

- **CLEVER score: Cross Lipschitz Extreme Value for nEtworK Robustness (CLEVER)** score, proposed by Reference [123], is a novel robustness evaluation metric. It is attack-independent and can be applied to any arbitrary neural network classifier and scales to large networks. CLEVER metric is an estimation of local Lipschitz constant, which represents “lower bound of the robustness in input data” or minimum amount of perturbation required to a natural sample to fail a classifier. The increased clever score indicates that the network is indeed made more resilient to adversarial perturbations after any defense mechanism is used.

9 ADVERSARIAL DATASETS AND FRAMEWORKS

There are several datasets in NLP that are proposed for adversarial evaluation. One such dataset is DailyDialog++ [98], which is an extension of DailyDialog dataset [69] and adversarial dialogue generation dataset. DailyDialog++ contains five additional relevant and adversarially irrelevant responses for 11k context conversation derived from DailyDialog. This dataset is used for evaluating

robustness of dialogue generation models with adversarial examples in the dataset against adversarial attacks. Further, these adversarial examples are used in adversarial training to improve the performance of dialogue generation models. Another work is ANLI [82], which proposed adversarial dataset for natural language inference systems. ANLI is composed of adversarial examples for NLI collected in three iterative rounds having human and machine in the loop. ANLI consists of 103k examples of sentences, starting with short multi-sentence passages from Wikipedia and having annotators write adversarial hypothesis. In this process, they tested these samples with state-of-the-art NLI models and got them verified by human annotators, hence proposing **human-and-model-in-the-loop enabled training (HAMLET)** scheme for data collection. The adversarial examples collected in this dataset are used in adversarial training to improve the performance of Natural Language Inferencing models.

In the same line, authors in Reference [66] proposed a novel large-scale dataset adversarial VQA for visual question answering task using the HAMLET scheme proposed in Reference [82]. In this work, they presented an image to an annotator and asked them to write a tricky question that could fool a model. Hence, they iteratively collected 243k questions for 37.9k images by having humans and models competing in the loop in three rounds. This dataset is used in evaluating the robustness of SOTA VQA models. Authors have further used this dataset for data augmenting for the purpose of adversarial training and demonstrated a higher performance on robust VQA benchmarks. For the purpose of evaluating adversarial examples for question answering task, authors in Reference [44] proposed Adversarial SQuAD dataset, which contained adversarially inserted sentences. These sentences are automatically generated in a concatenative manner, without changing the meaning of the paragraph or question. Fake answers to these questions are also generated with same POS type. This dataset is used for evaluating the performance of various state-of-the-art question answering models. The best-performing model with adversarial examples is analyzed for the features causing the high performance. Based on this analysis, additional features were incorporated in the model, separately and in combination, to enhance its performance both with and without adversarial examples. Authors further used this dataset for adversarial training of modified model to validate its stability. In the same line, Reference [109] proposed a question-answering test bed, Quizbowl, using a Human-In-the-Loop framework. In this work, human authors are asked to write adversarial questions that are designed to fail state-of-the-art question answering models appearing ordinary to human. This test dataset is used for adversarial evaluation of SOTA question answering models.

In another work [147], adversarial dataset, namely, **Paraphrase Adversaries from Word Scrambling (PAWS)**, for paraphrase detection, is proposed. PAWS is generated from sentence in Quora and Wikipedia, where adversarial samples are generated using language model-based controlled word swapping and back-translations. PAWS dataset is used for adversarial evaluation of paraphrase detection models and measured sensitivity of these models on word order and syntactic structure. Authors further used this dataset for adversarial training of state-of-the-art paraphrase detection models. In the direction of perturbation identification, Reference [127] proposed a dataset, **Text Classification and Attack Benchmark (TCAB)**, that was created for the purpose of detecting and labeling the textual perturbation in the input text. TCAB is an extensive dataset containing 1.5 million attack instances, generated using 12 attacks from the toolkits [80, 139] that are targeting three classifiers, trained on six domain datasets of sentiment and abuse classification. There is a total of 216 attacks taken into consideration in TCAB dataset, which is a combination of different types of attacks. This dataset consists of a total of 1, 53, 9, 881 adversarial examples along with the clean instances taken from the original datasets. They further use text, language, classifier properties to detect the perturbation and type of the attack in the input text. Table 9 shows the

Table 9. Summary of Adversarial Datasets in NLP

Dataset	Task	Statistics	Method
DailyDialog ++ [98]	Dialogue generation	11k context, 5 responses	Human Annotators
ANLI [82]	NLI systems	103k sentences	Human & machine in loop
Adversarial VQA [66]	Visual question answering	243k Q & 37.9k images	Human & machine in loop
Adversarial Squad [44]	Question answering	107,785 Q	Machine generated
Quizbowl [109]	Question answering	1,213 Q-A pairs	Human & machine in loop
PAWS [147]	Paraphrase detection	108,463 paraphrase pairs	Machine generated
TCAB [127]	Sentiment, Abuse classification	1.5M Adversarial ins.	Machine generated

Here NLI = Natural Language Inference, Q= Questions, A= Answers, ins= Instances.

summary of the publicly available adversarial datasets for various NLP tasks. It describes the task for which the dataset is proposed, its statistics and methods used in data collection and annotation.

There are papers in the literature that proposed Python frameworks for a complete adversarial evaluation for several NLP tasks with various attack algorithms. One such work is TextAttack [80], which is a Python framework for end-to-end adversarial evaluation with 16 different adversarial attack methods. It consists of a task-specific goal function, data augmentation schemes, along with perturbation constraints that validate the perturbation with original inputs and a repetitive model querying search system. It facilitates the user to benchmark existing attacks and create novel attack schemes by using new and existing components and evaluating them. Improving the shortcomings of TextAttack framework, Open attack [139] is proposed, which included 15 different types of attacks, such as sentence, word, character level. It also supports Chinese in addition to English language models and supports multi-process running of attack models to improve attack efficiency. In the same line, another evaluation framework [131], “Elephant in the room,” is proposed, which consists of a combination of automatic evaluation metrics and human judgments. Targeting the sentiment classification task, it included crowd-sourced human judgments for judging the naturalness, preservation of original label, and comparing similarity on a text similarity metric.

10 RECOMMENDATIONS FOR FUTURE WORK

In this article, an exhaustive survey of the methods proposed in the literature to defend neural networks from adversarial attacks and to enhance their robustness is presented. We proposed a novel taxonomy for adversarial defense mechanisms for various tasks in NLP. Methods for adversarial defense in NLP are broadly divided into three categories: (i) methods based on adversarial training, (ii) methods based on perturbation detection, (iii) methods providing a certificate for robustness. Another part of the methods proposed in literature that do not follow any of the above-mentioned schemes is categorized as miscellaneous. While there is an ample amount of work proposed in this direction for tasks in NLP, there are still various gaps remaining that should be looked at as potential future directions in this area.

- **Larger part of work based on adversarial training:** A large portion of work in adversarial defenses for NLP revolves around adversarial training and data augmentation. Despite having a plethora of work in this direction, there is a significant portion of adversarial defense methods that are oriented towards augmenting the data for generating adversarial examples for adversarial training. Adversarial training is undoubtedly a successful defense scheme for adversarial attacks, but it lacks generality for a more practical purpose. It makes the model highly robust for a certain kind of attack, but it still makes it vulnerable to the type of examples the model has not seen. Hence, the other methods for adversarial defenses should also be explored.

- **Hand-crafted generation of adversaries:** As a future work recommendation, more attention should be given to the automatic generation of adversarial examples. Most of the methods based on adversarial training with data augmentation rely on hand-crafted adversarial examples. There are methods that replace words with their synonyms or adjacent words, flip the character, or concatenate words at the end of sentences. Despite being highly efficient, these examples are devised by humans rather than having an automatic generation of adversarial examples and have their own limitations. Hence, more efforts can be put in the direction of the automatic generation of adversarial examples.
- **Interpretability of perturbations in text data:** Adversarial examples should be less human perceptible/natural looking to make the model robust in a practical attack scenario. In contrast with adversarial defenses for computer vision-based methods and examples generated on images, examples on text for NLP tasks are difficult to generate because of their discrete nature. Modifying pixels in images for generating adversarial examples are less perceptible to human than modifying a character or word in an input text. Hence, to make defense method more useful in a practical scenario, more efforts should be put in this direction.
- **Exact robust certificate calculation:** In existing literature, only the upper and lower bound on the certificate could be calculated, rather than the exact robustness certificate. While adversarial training-based defense methods provided a sufficient exposure towards achieving robust neural networks, progressively novel adversarial attacks kept rolling in. A new set of methods to achieve robustness was proposed in the direction of providing a certificate of robustness for a neural network, attempting to put an end to this race. Certification based adversarial defense methods, definitely provides a better generalization of defense against adversarial attacks on the neural network for a given task, but certification alone is not adequate for achieving robustness in a model. Finding an exact robustness certificate to the set of input is a non-convex optimization problem and is inefficient to solve. However, in the literature, authors have relaxed this problem to convex optimization by finding an upper or lower bound to the robustness certificate. Despite the efforts towards finding a certified robustness, convex optimization can lead to lossy results, and there is a scope of finding a tighter bound. Hence, future efforts in this direction can be made to improve the tightness of existing robustness certificates.
- **Scalability of certification-based robustness:** Certification methods do not scale to large and practical networks used in solving modern machine learning problems. The current certification-based robustness method in the literature is implemented on theoretical models on a small scale. They are not scalable to larger and deeper networks for practical purposes. Hence, in the future, attempts should be made in this direction.
- **Generalization of adversarial training:** The current state-of-the-art methods based on adversarial training in NLP are designed in a task-specific manner. There is a lack of generality in adversarial example generation schemes that could be used for multiple NLP tasks effectively. Therefore, steps can be taken in this direction in the future.
- **More explainable models in case of inserting perturbations in the loss function:** There is a lack of explainability and transparency in the regularization-based defense methods. The loss function contains the term responsible for introducing perturbations at the training time. However, there is no explanation for these methods for their correctness or high accuracy.
- **Interpretable adversarial training:** While adversarial training-based method covers a larger portion of adversarial defense literature in NLP, these models are hardly interpretable in terms of adversarial instances. Adversarial examples are provably helping these models

become more robust toward adversarial attacks, but there are questions that still need to be answered, such as: “How are they improving the performance?,” “Is this list of adversarial instances exhaustive or could there be more such instances?,” “Is the model robust towards some specific type of attacks or are they able to defend from any type of attacks?”.

- **Attack-agnostic perturbation detection:** A better part of the literature in perturbation detection methods works on the prior assumption of spelling-based attacks or synonym substitution-based attacks. However, adversarial perturbation in textual data could be caused by multiple types of attacks, individually or in combination. Hence, there is a requirement for perturbation detection schemes which are attack agnostic and presents a general framework for perturbation detection.
- **Novel methods to identify the existing perturbations in the input:** A large part of perturbation detection schemes depends upon spell-checking methods and methods that enumerate or cluster synonyms. In a practical scenario, it is highly inefficient to compute and enumerate synonyms of words for perturbation recognition. Hence, there is a requirement for novel and innovative methods for identifying the perturbations in the input text that do not involve traditional methods such as spell-checking, synonym mapping, or their other variants.
- **Better coverage for NLP applications:** The proposed adversarial defense methods evaluate their method on certain NLP tasks to validate the strength of their method. In addition to being limited to certain type adversarial attacks, a large part of defense methods demonstrate their algorithm on different types of text classification tasks. These tasks include sentiment, hate, news, topic, abuse, malware-classification problems which comes under the wider use-case of text classification. Text classification is one of the **Natural Language Understanding (NLU)** tasks, while there are other NLU tasks such as named entity recognition, machine translation, automatic reasoning, and so on, having lesser coverage. Moreover, a very limited number of defense schemes have demonstrated their methods on **Natural Language Generation (NLG)** use cases such as summary generation, question answering. Hence, there is a requirement for adversarial defense methods covering the other NLP applications.
- **Better evaluation metrics:** The current evaluation of robustness against adversarial attacks for NLP models is based on the performance metrics of the actual model, i.e., accuracy, precision-recall, error-analysis, and so on. Hence, there is a requirement for novel evaluation metrics that could measure the robustness and ability to defend against adversarial attacks. There is also a requirement for sensitivity metrics for machine learning models, for measuring their sensitivity towards adversarial examples. There also are not enough ways to evaluate defense mechanisms themselves along the lines of perceptibility and naturalness. Hence, in the future, more evaluation metrics can be brought along this line.

11 CONCLUSION

In this article, a survey is presented for adversarial defense methods for various tasks in NLP. We proposed a novel taxonomy for adversarial defenses in NLP covering a wide range of recently proposed papers. This survey tries to fulfill the gaps in existing surveys where adversarial attack schemes were more focused. However, in recent years, numerous methods have been proposed for defending neural networks with adversarial attacks and enhancing their robustness. Coming up with novel defense schemes for advanced NLP systems is as important as coming up with novel attacks to make these neural networks robust and safe for practical purposes. This survey also covers various adversarial datasets, frameworks proposed in recent times for efficient adversarial evaluation of the SOTA models, and evaluation metrics to quantify their robustness. Moreover, it highlights various recommendations for future work considering the limitations and gaps in

the existing literature on adversarial defenses. This survey, therefore, provides a strong basis and motivation for future research in developing robust and safe neural networks in NLP tasks.

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